



Normal safety valve with spring loading. (AN)

Mod. 285 - 286 | ASME | USCS



EP

AP

ES

EP

AP

ES

Mod. 285

Mod. 286

Operation

The valve works as an automatic pressure releasing regulator activated by the static pressure existing at the entrance to the valve and is characterized by its ability to open, at the first proportional to the pressure increase, and after instantly and totally.

Regulation

- PED 2014/68/EU
- UNE-EN ISO 4126-1
- UNE-EN 12516-2
- ASME B1.20.1
- ASME B16.5
- ASME VIII Div.1/API 520

Specifications

Size

- 1/2" x 1" to 1 1/4" x 2"

Temperature range

- -76 °F to +482 °F

Applications

- Gas, steam and liquids

Materials

- Cast steel
- Stainless steel

Maximum pressure

- Up to 363 psi

Certification



N°. PIECE	PIECE	MATERIAL	
		CAST STEEL	STAINLESS STEEL
1	Body	Cast steel (SA-216 Gr. WCB)	Stainless steel (SA-351 Gr. CF8M)
2	Bell	Cast steel (SA-216 Gr. WCB)	Stainless steel (SA-351 Gr. CF8M)
3	Seating	Stainless steel (SA-479 Gr. 316)	Stainless steel (SA-479 Gr. 316)
4	Lever	Stainless steel (AISI 304)	Stainless steel (AISI 304)
5	Hood	Cast steel (AISI 1045) (1)	Stainless steel (AISI 303) (2)
6	Hollow screw	Stainless steel (AISI 303)	Stainless steel (AISI 303)
7	Hollow scre nut	Stainless steel (AISI 303)	Stainless steel (AISI 303)
8	Rod	Stainless steel (AISI 316)	Stainless steel (AISI 316)
9	Ring (3)	Stainless steel (AISI 304)	Stainless steel (AISI 304)
10	Lead	Stainless steel (AISI 316)	Stainless steel (AISI 316)
11	Plug	Stainless steel (AISI 316)	Stainless steel (AISI 316)
12	Deflector	Stainless steel (AISI 316)	Stainless steel (AISI 316)
13	Stud	Stainless steel (AISI 316)	Stainless steel (AISI 316)
14	Sealing nut	Stainless steel (AISI 316)	Stainless steel (AISI 316)
15	Cap	Stainless steel (AISI 316)	Stainless steel (AISI 316)
16	Sealing wire	Hilo precintar	Hilo precintar
17	Safety ring (4)	Stainless steel (AISI 301)	Stainless steel (AISI 301)
18	Wire	Plastic	Plastic
19	Sealing disk	PTFE (Teflon)	PTFE (Teflon)
		Silicone rubber	Silicone rubber
		Fluorelastomer (Viton)	Fluorelastomer (Viton)
20	Spring press	Stainless steel (AISI 304)	Stainless steel (AISI 304)
21	Spring	Stainless steel (AISI 301)	Stainless steel (AISI 301)
22	Clip	Stainless steel (AISI 301)	Stainless steel (AISI 301)
23	Joint	PTFE (Teflon)	PTFE (Teflon)
24	Washer	PTFE (Teflon)	PTFE (Teflon)
25	Spring piston	Stainless steel (AISI 301)	Stainless steel (AISI 301)
26	Piston	Stainless steel (AISI 303)	Stainless steel (AISI 303)
27	Piston base	Stainless steel (AISI 303)	Stainless steel (AISI 303)
28	Separator	Stainless steel (AISI 304)	Stainless steel (AISI 304)
29	O-ring	Fluorelastomer (Viton)	Fluorelastomer (Viton)
NPT1 x NPT2		1/2" x 1" to 1 1/4" x 2"	
NPS1 x NPS2		1/2" x 1" to 1 1/4" x 2"	
CLASS		300 lbs	300 lbs
OPERATING CONDITIONS	PS [psig]	363	
	MAX. TEMP. [°F]	482	
	MIN. TEMP. [°F]	14	-76

(1) From NPT 1" x 1 1/2" / NPS 1" x 1 1/2" in Cast steel (SA-216 Gr. WCB)

(2) From NPT 1" x 1 1/2" / NPS 1" x 1 1/2" in Stainless steel (SA-351 Gr. CF8M)

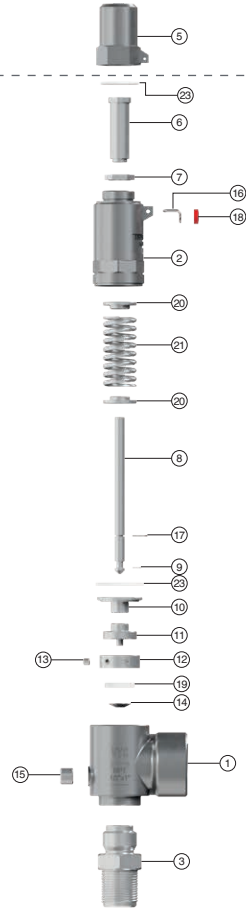
(3) Stainless steel (AISI 631) for NPT 1/2" x 1" / NPS 1/2" x 1" Stainless steel (AISI 301) for NPT 3/4" x 1 1/4" / NPS 3/4" x 1 1/4"

(4) From NPT 1" x 1 1/2" / NPS 1" x 1 1/2"

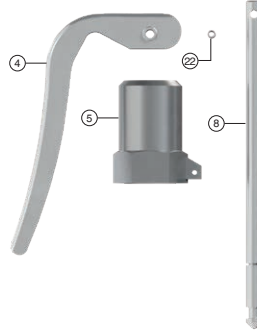
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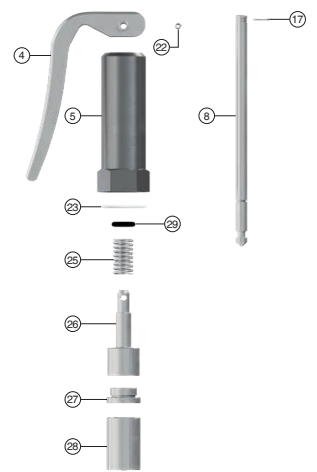
Isometric view ES



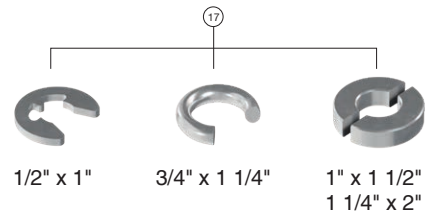
Exploded view ES



Exploded view AP



Exploded view EP

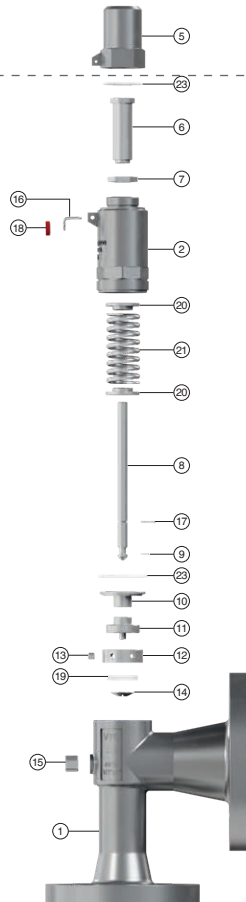


Rings

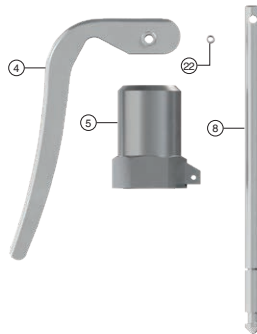
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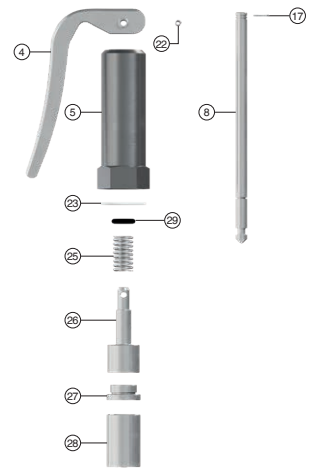
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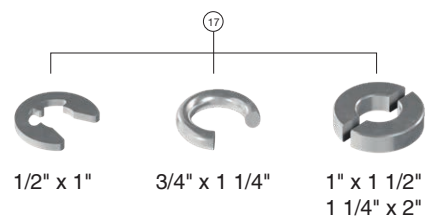
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Exploded view AP



Exploded view EP



Rings

NPT1 x NPT2			MODEL 285												
			1/2" x 1"			3/4" x 1 1/4"			1" x 1 1/2"			1 1/4" x 2"			
			Male thread NPT x Female thread NPT ASME B1.20.1												
NPS1 x NPS2			MODEL 286												
			1/2" x 1"			3/4" x 1 1/4"			1" x 1 1/2"			1 1/4" x 2"			
			Flanges ASME B16.5												
VERSION			ES	AP	EP	ES	AP	EP	ES	AP	EP	ES	AP	EP	
d0 [in]			0,59			0,59			0,79			1,10			
A0 [in2]			0,27			0,27			0,49			0,95			
(285) H [in]			6,65	7,01	8,62	8,66	9,17	10,51	10,63	11,14	12,36	10,63	11,14	8,43	
(286) H [in]			7,95	8,39	9,84	9,84	10,31	11,61	11,81	12,44	13,43	14,8	15,47	16,61	
h1 [in]			1,34			1,61			1,89			2,40			
(285) L1 [in]			1,65			1,89			2,13			2,68			
(285) L2 [in]			2,32			2,64			3,07			3,58			
(286) L1 [in]			3,15			3,35			3,74			3,94			
(286) L2 [in]			3,54			3,74			4,13			4,33			
R			1/8"			1/8"			1/8"			1/8"			
			Female threaded Gas Whitworth cylindrical UNE-EN ISO 228-1												
INLET FLANGE	CLASS 300 ASME B16.5	D1 [in]	3,75			4,62			4,88			5,25			
		K1 [in]	2,62			3,25			3,50			3,88			
		d1 [in]	5/8			3/4			3/4			3/4			
		C1 [in]	0,56			0,62			0,69			0,75			
		DRILLS N°	4			4			4			4			
OUTLET FLANGE	CLASS 150 ASME B16.5	D2 [in]	4,25			4,62			5,00			6,00			
		K2 [in]	3,12			3,50			3,88			4,75			
		d2 [in]	5/8			5/8			5/8			3/4			
		C2 [in]	0,56			0,62			0,69			0,75			
		DRILLS N°	4			4			4			4			
WEIGHT [lb]		(285) CAST STEEL	1,5	1,6	1,8	2,2	2,3	2,6	5,2	5,4	6,0	10,3	10,5	11,3	
		(285) STAINLESS STEEL													
		(286) CAST STEEL	5,9	6,0	7,6	9,8	9,8	10,9	13,7	14,0	15,2	21,0	21,6	23,8	
		(286) STAINLESS STEEL													
CODE		CAST STEEL 2002-285.	60242	60241	60243	63442	63441	63443	61042	61041	61043	61442	61441	61443	
		STAINLESS STEEL 2002-285.	60222	60221	60223	63422	63421	63423	61022	61021	61023	61422	61421	61423	
		CAST STEEL 2002-286.	60242	60241	60243	63442	63441	63443	61042	61041	61043	61442	61441	61443	
		STAINLESS STEEL 2002-286.	60222	60221	60223	63422	63421	63423	61022	61021	61023	61422	61421	61423	
285															
286															

PRESSURES AND SPRING REGULATION RANGES						
NPT1 x NPT2			1/2" x 1"	3/4" x 1 1/4"	1" x 1 1/2"	1 1/4" x 2"
NPS1 x NPS2			1/2" x 1"	3/4" x 1 1/4"	1" x 1 1/2"	1 1/4" x 2"
SET PRESSURE [psig]	MAXIMUM (LIQUIDS AND GASES)		363	363	363	363
	MAXIMUM (SATURATED STEAM)		363	363	363	363
	MINIMUM	STEAM AND GASES	7,3	7,3	7,3	7,3
		LIQUIDS	2,9	2,9	2,9	2,9
SPRING REGULATION RANGES [psig]	2,9 - 10,2	CODE	56341	56348	56356	56364
	7,3 - 23,2	CODE	56342	56349	56357	56365
	20,3 - 79,8	CODE	56343	56350	56358	56366
	43,5 - 50,8	CODE	56344	56351	56359	56367
	72,5 - 145,0	CODE	56345	56352	56360	56368
	130,5 - 290,1	CODE	56346	56353	56361	56369
	203,1 - 290,1	CODE		56354	56362	56370
	275,6 - 363,0	CODE	56347	56355	56363	56371
DISCHARGE COEFFICIENTS						
NPT1 x NPT2			1/2" x 1"	3/4" x 1 1/4"	1" x 1 1/2"	1 1/4" x 2"
NPS1 x NPS2			1/2" x 1"	3/4" x 1 1/4"	1" x 1 1/2"	1 1/4" x 2"
d0 [in]			0,59	0,59	0,79	1,10
h [in]			0,09	0,15	0,20	0,28
h/d0 [in]			0,0055	0,0098	0,0098	0,0098
STEAM AND GAS [Kdr]			0,35	0,62	0,62	0,62
LIQUIDS [Kdr]			0,27	0,54	0,54	0,54



Fluid dynamics

Depending on demand

1.- Silicone's rubber, Fluorelastomer (Viton) seals, PTFE (Teflon)... etc., achieving leakage levels less than:

$$0,2 \times 10^{-8} \frac{\text{psi pulg.}^3}{\text{seg.}}$$

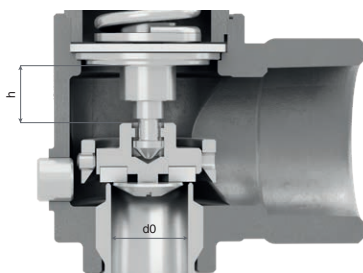
The ranges of application allow certain flexibility although we recommend limiting them to:

TEMPERATURE RANGE IN °F OF THE SEALS			
SEALS		MINIMUM	MAXIMUM
SILICONE RUBBER	S	-76	392
FLUORELASTOMER (VITON)	V	-4	392
PTFE (TEFLON)	T	-328	500

RECOMMENDED RANGES OF APPLICATION			
VERSION	AP	ES	EP
FLUID	SATURATED STEAM	* (1)	*
	GAS	* (1)	*
	LIQUID	*	*

(1) If working with expensive or harmful fluids, use only ES or EP models.

RECOMMENDED RANGES OF APPLICATION				
NPT1 x NPT2	1/2" x 1"	3/4" x 1 1/4"	1" x 1 1/2"	1 1/4" x 2"
NPS1 x NPS2	1/2" x 1"	3/4" x 1 1/4"	1" x 1 1/2"	1 1/4" x 2"
d0 [in]	0,59	0,59	0,79	1,10
h [in]	0,09	0,15	0,20	0,28
h/d0 [in]	0,0055	0,0098	0,0098	0,0098
STEAM AND GAS [Kdr]	0,35	0,62	0,62	0,62
LIQUID [Kdr]	0,27	0,54	0,54	0,54



OPENING, OVERPRESSURE AND RESET PRESSURES			
FLUID	OPENING PRESSURE	OVERPRESSURE	RESET PRESSURE
SATURATED STEAM AND GASES	The higher of ± 3% or ± 1,45 psi	The higher of ± 3% or ± 1,45 psi	The higher of ± 15% or 4,35 psi
LIQUIDS	The higher of ± 3% or ± 1,45 psi	The higher of ± 3% or ± 1,45 psi	The higher of ± 20% or 8,70 psi

Specifications

- 90° angular flow.
- Activated by direct action helicoid spring.
- Simplicity of construction ensuring minimum maintenance.
- Materials carefully selected for their resistance to corrosion.
- Internal body designed to offer favourable flow profile.
- Seat and sealing disk balanced, making them extremely tightness, even exceeding EN 12266-1.
- Great discharge capacity.
- Deflector nut designed to make easier the steam expansion, a sudden opening and to measure the blowdown of any fluid.
- Guarantees absolute opening and closing precision.
- Equipped with draining screws for removing condensation.
- Orientation of the lever by rotation.
- All the valves are supplied sealed at the set pressure requested, simulating operational conditions, and are vigorously tested.
- All components are numbered, registered and checked. If requested in advance, material, casting, test and efficiency certificates will be enclosed with the valve, and the instruction manual, in accordance with P.E.D. 2014/68/EU.

CAPACIDADES DE DESCARGA												
NPT1 x NPT2	1/2" x 1"			3/4" x 1 1/4"			1" x 1 1/2"			1 1/4" x 2"		
NPS1 x NPS2	1/2" x 1"			3/4" x 1 1/4"			1" x 1 1/2"			1 1/4" x 2"		
d0 [mm]	0,59			0,59			0,79			1,1		
A0 [mm2]	0,27			0,27			0,49			0,95		
p [psig]	I	II	III	I	II	III	I	II	III	I	II	III
14,5	142	55	11	251	97	22	446	173	40	875	339	78
21,8	179	69	14	317	123	28	563	218	49	1104	427	96
29,0	216	84	16	383	148	32	681	263	57	1334	516	111
36,3	253	98	18	449	173	36	798	308	63	1563	605	124
43,5	290	112	19	515	199	39	915	354	69	1793	693	136
50,8	328	127	21	581	224	42	1032	399	75	2023	782	147
58,0	365	141	22	647	250	45	1149	444	80	2253	871	157
65,3	402	155	24	712	275	48	1267	489	85	2483	959	166
72,5	439	170	25	779	301	50	1384	535	89	2713	1048	175
79,8	477	184	26	845	326	53	1501	580	94	2943	1136	184
87,0	514	198	28	911	352	55	1619	625	98	3173	1225	192
94,3	551	213	29	977	377	57	1736	670	102	3403	1314	200
101,5	589	227	30	1043	402	59	1854	716	106	3634	1402	207
108,8	626	242	31	1109	428	62	1972	761	109	3864	1491	215
116,0	663	256	32	1175	453	64	2089	806	113	4095	1580	222
130,5	738	285	34	1308	504	67	2325	896	120	4557	1757	235
145,0	813	313	36	1440	555	71	2561	987	126	5019	1934	248
159,5	888	342	37	1573	606	75	2797	1077	133	5481	2112	260
174,0	963	371	39	1706	657	78	3033	1168	138	5944	2289	271
188,5	1038	400	41	1839	708	81	3269	1258	144	6407	2466	283
203,0	1113	428	42	1972	759	84	3506	1349	150	6871	2644	293
217,5	1188	457	44	2105	810	87	3742	1439	155	7335	2821	303
232,0	1264	486	45	2238	860	90	3979	1530	160	7800	2998	313
246,5	1339	514	46	2372	911	93	4217	1620	165	8265	3176	323
261,0	1414	543	48	2505	962	95	4454	1711	170	8730	3353	332
275,5	1490	572	49	2639	1013	98	4692	1801	174	9196	3530	342
290,0	1565	601	50	2773	1064	101	4930	1892	179	9663	3708	350
304,5	1641	629	52	2907	1115	103	5168	1982	183	10129	3885	359
319,0	1717	658	53	3041	1166	105	5407	2073	188	10597	4062	368
333,5	1793	687	54	3175	1217	108	5645	2163	192	11065	4240	376
348,0	1868	716	55	3310	1268	110	5884	2254	196	11533	4417	384
362,5	1944	744	56	3444	1318	112	6123	2344	200	12002	4594	392

I - Saturated steam [lb/h].

II - Air at 60 °F and 14,50 psi [S.C.F.M.].

III - Water at 70 °F [US - G.P.M.].

ATTENTION: Flow rates according to ASME VIII Div.1/API 520 with 10% overpressure.



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